

## Customer Journey Map

|                      |                 |                       |              |
|----------------------|-----------------|-----------------------|--------------|
| <b>Date:</b>         | 08 October 2026 | <b>Team ID:</b>       | UNM1611JASON |
| <b>Project Name:</b> | AI BlogNest API | <b>Maximum Marks:</b> | 2 Marks      |

### Customer Experience Stage-by-Stage Mapping:

Map out the customer's end-to-end experience across three core interaction phases, capturing their actions, touchpoints, thoughts, feelings, process ownership, and improvement opportunities.

| Phase of Journey                                     | Stage 1: Onboarding & Auth                                                                                                                                                                             | Stage 2: AI Generation                                                                                                                                                                                  | Stage 3: Blog Management                                                                                                                                                                       |
|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actions</b><br>What does the user do?             | <ul style="list-style-type: none"> <li>Registers account via /register</li> <li>Logs in with email &amp; password</li> <li>Acquires Bearer JWT token</li> <li>Inspects profile via /profile</li> </ul> | <ul style="list-style-type: none"> <li>Submits prompt topic &amp; category</li> <li>Triggers /generate-blog</li> <li>Submits raw text to /summarize</li> <li>Reviews auto-categorized output</li> </ul> | <ul style="list-style-type: none"> <li>Creates manual blog posts</li> <li>Browses all published articles</li> <li>Views individual blog by ID</li> <li>Updates or deletes own posts</li> </ul> |
| <b>Touchpoints</b><br>What service part do they use? | <ul style="list-style-type: none"> <li>Thunder Client / Postman Client</li> <li>Express Auth Routes</li> <li>Bcrypt hashing module</li> <li>MongoDB Users Collection</li> </ul>                        | <ul style="list-style-type: none"> <li>POST /api/ai/generate-blog</li> <li>POST /api/ai/summarize</li> <li>Google Gemini Generative AI SDK</li> <li>Text sanitization &amp; regex engine</li> </ul>     | <ul style="list-style-type: none"> <li>/api/blogs CRUD endpoints</li> <li>Auth verification middleware</li> <li>Mongoose query &amp; populate</li> <li>MongoDB Blogs Collection</li> </ul>     |
| <b>Customer Thoughts</b><br>What are they thinking?  | <ul style="list-style-type: none"> <li>"Is my password securely encrypted?"</li> <li>"Will the JWT token persist my session?"</li> <li>"Is the auth response format clear?"</li> </ul>                 | <ul style="list-style-type: none"> <li>"Will Gemini return a complete post?"</li> <li>"Is the generation speed fast enough?"</li> <li>"Did it categorize my topic accurately?"</li> </ul>               | <ul style="list-style-type: none"> <li>"Can other users modify my blog post?"</li> <li>"How quickly does the blog list load?"</li> <li>"Are populated author details accurate?"</li> </ul>     |
| <b>Customer Feeling</b><br>What are they feeling?    | <ul style="list-style-type: none"> <li>Reassured by rapid token generation</li> <li>Protected by cryptographic hashing</li> </ul>                                                                      | <ul style="list-style-type: none"> <li>Amazed by high-quality AI text</li> <li>Relieved from tedious drafting tasks</li> </ul>                                                                          | <ul style="list-style-type: none"> <li>In full control of their blog catalog</li> <li>Confident in strict author ownership</li> </ul>                                                          |

## Customer Journey Map (Continued) — Governance & Opportunities

| Phase of Journey                                        | Stage 1: Onboarding & Auth                                                                                                                                                                                                 | Stage 2: AI Generation                                                                                                                                                                                                      | Stage 3: Blog Management                                                                                                                                                                                                                       |
|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Process Ownership</b><br>Who is in the lead on this? | <b>Backend Security Lead:</b><br>Manages JWT generation, token validation, password salting, and credential security.                                                                                                      | <b>AI / ML Lead:</b><br>Monitors Gemini API latency, tunes prompt constraints, optimizes regex cleaning & categorization.                                                                                                   | <b>Database &amp; API Lead:</b><br>Maintains Mongoose models, optimizes query indexes, and enforces author authorization guards.                                                                                                               |
| <b>Opportunities</b><br>How can we improve this stage?  | <ul style="list-style-type: none"> <li>• Implement OAuth2 / Google SSO</li> <li>• Add email verification workflow</li> <li>• Refresh token rotation mechanism</li> <li>• Role-based access control (Admin/User)</li> </ul> | <ul style="list-style-type: none"> <li>• Add streaming SSE responses</li> <li>• Support custom word length parameter</li> <li>• Multi-language blog translation</li> <li>• Tone selection (Formal, Casual, Tech)</li> </ul> | <ul style="list-style-type: none"> <li>• Add full-text search with MongoDB Atlas</li> <li>• Implement pagination &amp; tag filtering</li> <li>• Redis caching for popular blogs</li> <li>• Markdown &amp; rich-text editor frontend</li> </ul> |

### Journey Experience Metrics & Impact:

| Metric                      | Baseline / Manual | AI BlogNest API           | Gain                    |
|-----------------------------|-------------------|---------------------------|-------------------------|
| Time to draft 150-word post | 30 - 45 minutes   | 1.8 seconds               | 96% time saved          |
| Time to summarize article   | 10 - 15 minutes   | 1.4 seconds               | 90% time saved          |
| User onboarding to API call | 15+ minutes       | < 2 minutes               | Turnkey REST JSON       |
| Author ownership protection | Manual DB checks  | 100% automated middleware | Zero unauthorized edits |